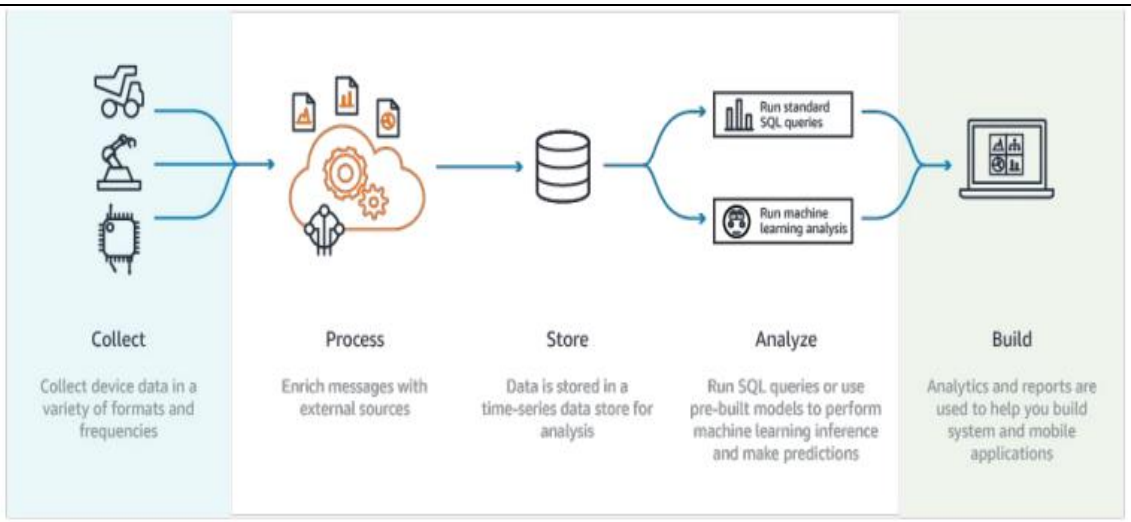
IoT Cloud Processing & Analytics - AWS IoT Core and Dynamo DB

IoT Data Lifecycle



- Data Collection
- Data Transmission
- Data Storing
- Data Processing Real Time & Batch
- Reporting & Visualization

AWS IoT Services

Device & Edge Services

Greengrass :- SDK for Edge Servers

FreeRTOS - Real-time OS for Devices

FreeRTOS BLE Mobile SDK - Low power communication via mobiles

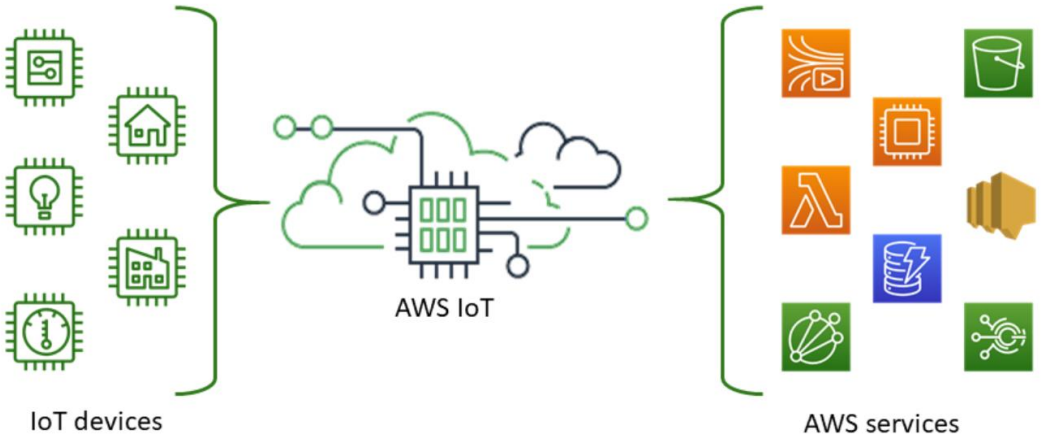
Cloud Service for IoT

- **IoT Core** - Scalable and secure device connection and communication, Device shadows, On-the-fly data transformation, post-collection seamless connection to various AWS services
- **IoT Device Management -** Registration, Organization, Monitoring, Update Management

Analytics Service

- **AWS IoT Analytics** - Workflows, Time-series data storage, Queries, Transformations, ML
- **IoT Events** - Rule-based event processing and actions
- **IoT Things Graph** - Visual organization and workflows

AWS IoT Core



https://docs.aws.amazon.com/iot/latest/developerguide/what-is-aws-iot.html
https://aws.amazon.com/iot-core/

AWS-IOT

- Managed IoT services by AWS
- Connect and manage billions of devices.
- Bridge between Devices & AWS Services
- Collect, store, & analyse IoT data

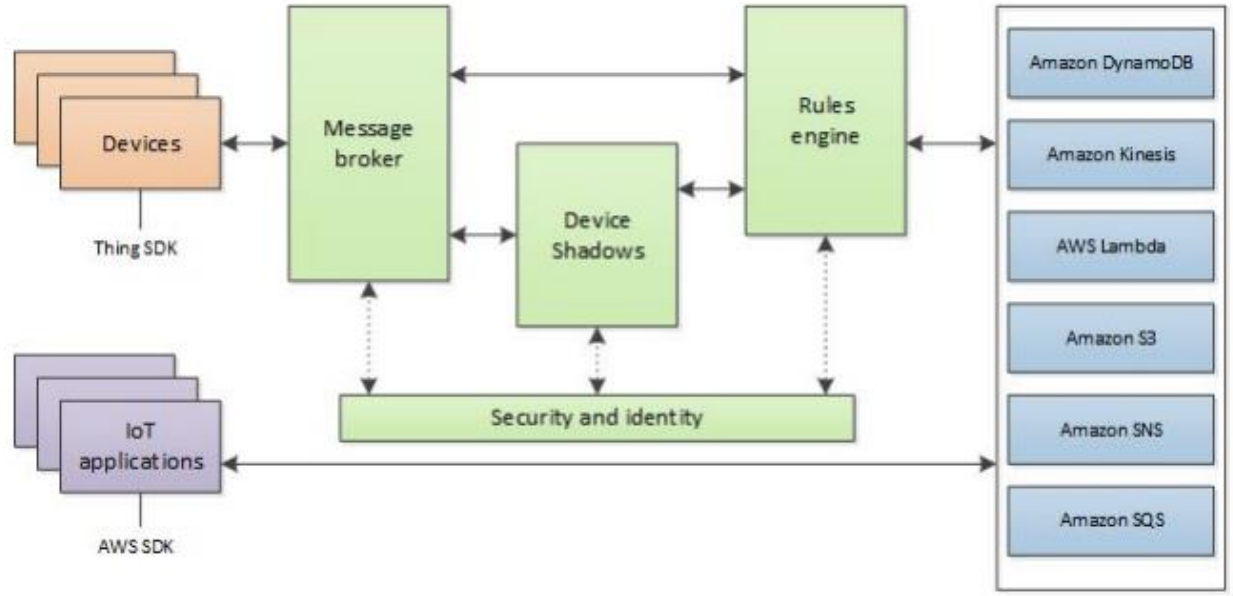
Similar Service with Azure & GCP

- Azure IoT Hub
- Google Cloud IoT Core

IOT Core

- Virtual device configuration and mapping to real devices
- Shadow device service to maintain status, monitoring, updates, and upstream connections
- Device organization in various hierarchical categories
- Secure device communication - X.509 certificate and keys for authentication and authorization
- Communication protocols - MQTTS, MQTT over secure websockets, HTTPS, LoRaWAN
- On-the-fly data transformation
- Data streaming to various services - DynamoDB, S3, SNS, Lambda, Elasticsearch, Cloudwatch, IoT Analytics, IoT Events, Kinesis, SQS, Custom services
- Scalable to billions of devices

AWS IOT Core Components



Thing	• Is the the basic building block. Thing. A Thing can be an end device generating sensor data using one or more sensors. Edge devices using Greengrass SDK can be configured as a Greengrass core device.
IoT Device Gateway & Message broker	• Multiple Communication protocols - MQTTS, MQTT over secure websockets, HTTPS, LoRaWAN • Low latency scaling to billions of devices • Bidi communication with devices - data ingress, heartbeat monitoring, control messages, updates
IoT Device Registry	• Virtual device registration and Bulk provisioning with hierarchies (using API) • Secure communication automated API-driven provisioning & management

IoT Device Shadow	• Additionally, you can also set the initial parameters that should be part of the device shadow.
	  • State management with continuous updates
	  • Bidi communication for downstream configuration and software updates
	  • Status available to upstream services even on device disconnections
	  • Device state restoration on reconnection and supports generic Get, Update , Delete REST api calls
	https://docs.aws.amazon.com/iot/latest/developerguide/iot-device-shadows.html

Jobs Management and API	• Defined jobs for devices - OS updates, software installs, reboot, cache reset, troubleshooting information, security certificate updates • Extensive control over device targeting and job ordering • Support for supplying assets including new updated files and configuration information

IoT Rule Engine	• Enables automatic upstream data and event communication to various services - DynamoDB, S3, SNS, Lambda, Elasticsearch, Cloudwatch, IoT Analytics, IoT Events, Kinesis, SQS, Custom services
	• SQL based selection syntax for
	  ○ Partial payload pickup possible - select value, device_id, timestamp
	  ○ Topic selection control - from devices/# . Possible data pickup or alerts, based on rules - where value >= 50
	• Use cases - Storage, Stream Processing, Direct alerts/notifications, Processing triggers based on conditions, etc.

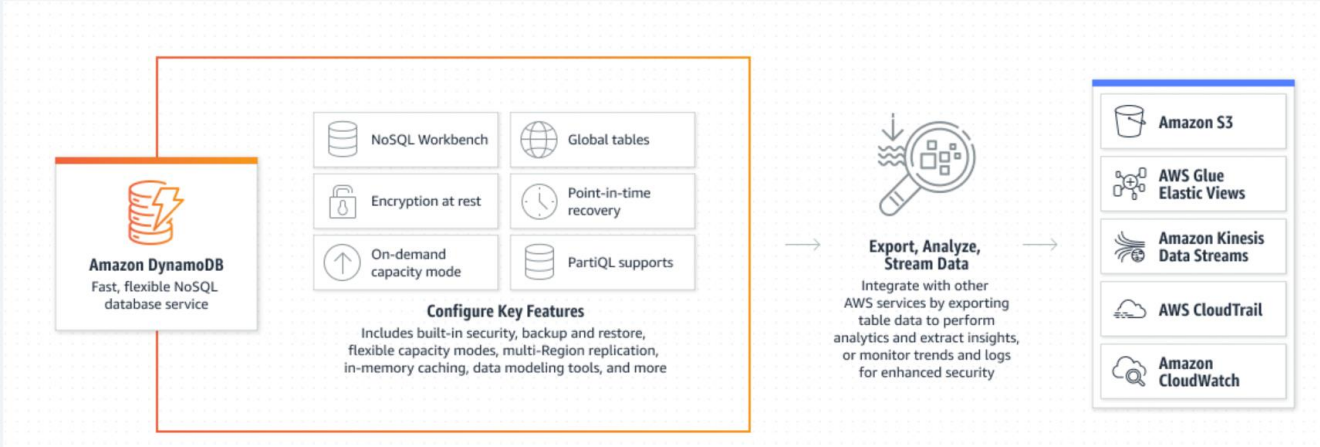
DynamoDB action rule	• Automatically send all (or some) device data to DynamoDB . Map different set of topics to different tables for automatic segregation. Complete payload in one column or map payload keys to different columns

AWS Dynamo DB

NoSQL DB :- https://www.mongodb.com/resources/basics/databases/nosql-explained

AWS Dynamo DB - https://aws.amazon.com/dynamodb/

Dynamo VS Document DB - https://dynobase.dev/dynamodb-vs-documentdb/



- **Fully Managed Serverless NoSQL DB** by AWS – No installation, upgrade, capacity planning, or manual provisioning needed
  - Auto Scalable with simple rules based on size and read/write load
  - Redundant - No single point of failure, muti-zone & multi-region
  - High Performance with nearly unlimited throughput and storage
  - Very low latency – as it has hash maps – fast retrieval of data
  - Secure transmission, encrypted at rest, Backup/Restore
- **NoSQL data model**:-
  - Collection (tables in RDBMS)
  - items (row in RDMBS) ;
  - items has attributes (equivalent to column); supports nested attribute up to 32 levels
  - Access based optimization - Partition and Sort keys for efficient single or range pickup
- **Ingress / Egress**:-
  - Ingress - Simple ingestion setup from IoT Core and other services
  - Egress - Base for batch processing and analytics services, item-level change streaming to Kinesis

References

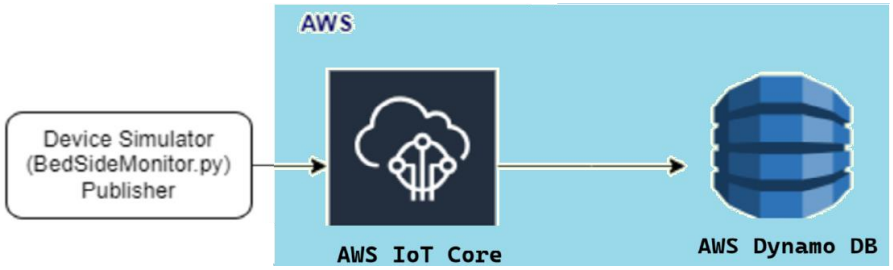
https://docs.aws.amazon.com/iot/latest/developerguide/what-is-aws-iot.html
https://aws.amazon.com/iot-core/
https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/Introduction.html

Hands On

Practice Material Provided

1. Step by Step guide - 4.1-Main-IoT-Core-BedSideMonitor-Example
2. BedSideMonitor.py – Python Source code by Device Simulator

Component Diagram



AWS IoT Core Components	BedsideMonitor.py (python code for device simulator)	Dynamo DB

• Thing
  • Name, Group, Type
  • Policy
• Device Shadow
• MQTT Test Client
• Dynamo DB
• Role
• Thing Rule to insert incoming data in MQTT into DB
	• For connecting to AWS Core
	  • End point of Thing
	  • Security certificates
	  • Topic
	• Python library
	  • AWSIoTPythonSDK
	  • Boto3
		• Role
		• Collection
		•